

tables

November 26, 2025

```
[40]: import pandas as pd
      from pathlib import Path

[41]: def load_results_table(results_dir, methods_list, use_coords, use_demographic,
      ↪file_name="aggregated_results.json"):
      agg_list = []
      for method in methods_list:
          if use_demographic and use_coords:
              folder_name = f"{method}_coords_demographic_radiomics"
          elif use_demographic and not use_coords:
              folder_name = f"{method}_demographic_radiomics"
          elif not use_demographic and use_coords:
              folder_name = f"{method}_coords_radiomics"
          else:
              folder_name = f"{method}_radiomics"
          aggregated_result_dir = Path(results_dir)/folder_name / file_name
          df = pd.read_json(aggregated_result_dir)
          agg_list.append(df)
      final_df = pd.concat(agg_list, ignore_index=True)
      final_df = final_df[['model', 'Accuracy_Mean', 'ROC_AUC_Mean']]
      display(final_df)
      print("Average across models:")
      display(pd.DataFrame(final_df.mean(numeric_only=True)).T)
```

1 Five Fold Cross Validation for Center 1 Dataset

```
[42]: print("Radiomics Features:")
      load_results_table("../Results", ["transformer", "mil", "deep_sets", "graph",
      ↪"ml_models"], use_coords=False, use_demographic=False)
```

Radiomics Features:

	model	Accuracy_Mean	ROC_AUC_Mean
0	transformer	0.730931	0.749828
1	mil	0.738027	0.743428
2	deep_sets	0.718257	0.755376
3	graph	0.664220	0.679778

4	AdaBoost	0.711806	0.733495
5	CatBoost	0.732265	0.747641
6	Extra Trees	0.704694	0.762465
7	Gradient Boosting	0.697353	0.728992
8	Logistic Regression	0.717827	0.709179
9	MLP	0.717597	0.759244
10	Random Forest	0.691576	0.770567
11	SVM (RBF kernel)	0.677783	0.666634
12	XGBoost	0.691806	0.696084
13	k-Nearest Neighbors	0.710011	0.737648
14	lightGBM	0.691806	0.695400

Average across models:

	Accuracy_Mean	ROC_AUC_Mean
0	0.706397	0.72905

```
[43]: print("Radiomics + demographic Features:")
load_results_table("../Results", ["transformer", "mil", "deep_sets", "graph",
↪ "ml_models"], use_coords=False, use_demographic=True)
```

Radiomics + demographic Features:

	model	Accuracy_Mean	ROC_AUC_Mean
0	transformer	0.751175	0.866042
1	mil	0.784509	0.860924
2	deep_sets	0.764279	0.837169
3	graph	0.764509	0.837384
4	AdaBoost	0.703789	0.859971
5	CatBoost	0.751376	0.852510
6	Extra Trees	0.717812	0.822527
7	Gradient Boosting	0.750931	0.848707
8	Logistic Regression	0.712050	0.704837
9	MLP	0.757842	0.833635
10	Random Forest	0.743604	0.829401
11	SVM (RBF kernel)	0.677783	0.666634
12	XGBoost	0.703804	0.828690
13	k-Nearest Neighbors	0.710011	0.737648
14	lightGBM	0.657093	0.794990

Average across models:

	Accuracy_Mean	ROC_AUC_Mean
0	0.730038	0.812071

```
[44]: print("Radiomics + coords Features:")
load_results_table("../Results", ["transformer", "mil", "deep_sets", "graph",
↪ "ml_models"], use_coords=True, use_demographic=False)
```

Radiomics + coords Features:

	model	Accuracy_Mean	ROC_AUC_Mean
0	transformer	0.812095	0.882030
1	mil	0.791175	0.835725
2	deep_sets	0.791850	0.860978
3	graph	0.751850	0.776708
4	AdaBoost	0.738717	0.765961
5	CatBoost	0.745384	0.792071
6	Extra Trees	0.738717	0.800776
7	Gradient Boosting	0.744279	0.766537
8	Logistic Regression	0.704709	0.696208
9	MLP	0.764294	0.813629
10	Random Forest	0.731591	0.794877
11	SVM (RBF kernel)	0.677783	0.666634
12	XGBoost	0.698702	0.754552
13	k-Nearest Neighbors	0.710011	0.737648
14	lightGBM	0.725384	0.774504

Average across models:

	Accuracy_Mean	ROC_AUC_Mean
0	0.741769	0.781256

```
[45]: print("Radiomics + coords + demographic Features:")
load_results_table("../Results", ["transformer", "mil", "deep_sets", "graph",
↪ "ml_models"], use_coords=True, use_demographic=True)
```

Radiomics + coords + demographic Features:

	model	Accuracy_Mean	ROC_AUC_Mean
0	transformer	0.818547	0.910149
1	mil	0.811405	0.903351
2	deep_sets	0.825643	0.909966
3	graph	0.840341	0.887535
4	AdaBoost	0.731146	0.858500
5	CatBoost	0.737382	0.850022
6	Extra Trees	0.738042	0.848018
7	Gradient Boosting	0.750486	0.849957
8	Logistic Regression	0.704709	0.698761
9	MLP	0.778072	0.855365
10	Random Forest	0.751376	0.847382
11	SVM (RBF kernel)	0.677783	0.666634
12	XGBoost	0.703804	0.827850
13	k-Nearest Neighbors	0.710011	0.737648
14	lightGBM	0.656878	0.802316

Average across models:

	Accuracy_Mean	ROC_AUC_Mean
0	0.749042	0.83023

2 Second dataset

```
[46]: print("Radiomics Features:")
load_results_table("../Results", ["transformer", "mil", "deep_sets", "graph",
    ↪ "ml_models"], use_coords=False, use_demographic=False,
    ↪ file_name="center2_aggregated_results.json")
```

Radiomics Features:

	model	Accuracy_Mean	ROC_AUC_Mean
0	transformer	0.6225	0.655303
1	mil	0.5725	0.602652
2	deep_sets	0.5400	0.580871
3	graph	0.5475	0.586995
4	AdaBoost	0.6325	0.666856
5	CatBoost	0.6500	0.682576
6	Extra Trees	0.6325	0.670455
7	Gradient Boosting	0.6225	0.636111
8	Logistic Regression	0.5825	0.641414
9	MLP	0.5875	0.642109
10	Random Forest	0.6275	0.675505
11	SVM (RBF kernel)	0.6250	0.726389
12	XGBoost	0.6350	0.683649
13	k-Nearest Neighbors	0.6350	0.661679
14	lightGBM	0.6175	0.652525

Average across models:

	Accuracy_Mean	ROC_AUC_Mean
0	0.608667	0.651006

```
[47]: print("Radiomics + demographic Features:")
load_results_table("../Results", ["transformer", "mil", "deep_sets", "graph",
    ↪ "ml_models"], use_coords=False, use_demographic=True,
    ↪ file_name="center2_aggregated_results.json")
```

Radiomics + demographic Features:

	model	Accuracy_Mean	ROC_AUC_Mean
0	transformer	0.6625	0.787879
1	mil	0.6300	0.699495
2	deep_sets	0.6225	0.639899
3	graph	0.5900	0.685101
4	AdaBoost	0.6800	0.848485
5	CatBoost	0.7125	0.847980
6	Extra Trees	0.7175	0.769949
7	Gradient Boosting	0.6650	0.797096
8	Logistic Regression	0.5875	0.634722
9	MLP	0.6350	0.671402
10	Random Forest	0.7000	0.796023

11	SVM (RBF kernel)	0.6250	0.726389
12	XGBoost	0.6850	0.844571
13	k-Nearest Neighbors	0.6350	0.661679
14	lightGBM	0.6900	0.822664

Average across models:

	Accuracy_Mean	ROC_AUC_Mean
0	0.655833	0.748889

```
[48]: print("Radiomics + coords Features:")
load_results_table("../Results", ["transformer", "mil", "deep_sets", "graph",
↪ "ml_models"], use_coords=True, use_demographic=False,
↪ file_name="center2_aggregated_results.json")
```

Radiomics + coords Features:

	model	Accuracy_Mean	ROC_AUC_Mean
0	transformer	0.6200	0.708712
1	mil	0.6150	0.635101
2	deep_sets	0.5400	0.534154
3	graph	0.6325	0.670328
4	AdaBoost	0.6400	0.728030
5	CatBoost	0.6600	0.710606
6	Extra Trees	0.6375	0.675189
7	Gradient Boosting	0.6775	0.715278
8	Logistic Regression	0.5825	0.644444
9	MLP	0.5950	0.646780
10	Random Forest	0.6625	0.694318
11	SVM (RBF kernel)	0.6250	0.726389
12	XGBoost	0.6500	0.731124
13	k-Nearest Neighbors	0.6350	0.661679
14	lightGBM	0.6475	0.728283

Average across models:

	Accuracy_Mean	ROC_AUC_Mean
0	0.628	0.680694

```
[49]: print("Radiomics + demographic + coords Features:")
load_results_table("../Results", ["transformer", "mil", "deep_sets", "graph",
↪ "ml_models"], use_coords=True, use_demographic=True,
↪ file_name="center2_aggregated_results.json")
```

Radiomics + demographic + coords Features:

	model	Accuracy_Mean	ROC_AUC_Mean
0	transformer	0.7075	0.811364
1	mil	0.6700	0.741793
2	deep_sets	0.6775	0.753030
3	graph	0.7150	0.794634

4	AdaBoost	0.6725	0.854167
5	CatBoost	0.7175	0.843687
6	Extra Trees	0.6975	0.782702
7	Gradient Boosting	0.7325	0.825000
8	Logistic Regression	0.5875	0.632955
9	MLP	0.6625	0.739078
10	Random Forest	0.7250	0.817740
11	SVM (RBF kernel)	0.6250	0.726389
12	XGBoost	0.6850	0.845960
13	k-Nearest Neighbors	0.6350	0.661679
14	lightGBM	0.7025	0.831755

Average across models:

	Accuracy_Mean	ROC_AUC_Mean
0	0.680833	0.777462

3 Test Time Adaptation (TTA) on Center 2 data

3.1 TENT method for TTA

```
[50]: print("Radiomics + demographic + coords Features:")
load_results_table("../Results", ["transformer", "mil", "deep_sets", "graph"],
    use_coords=True, use_demographic=True,
    file_name="tta_tent_center2_aggregated_results.json")
```

Radiomics + demographic + coords Features:

	model	Accuracy_Mean	ROC_AUC_Mean
0	transformer	0.7075	0.868182
1	mil	0.7700	0.864141
2	deep_sets	0.7725	0.857702
3	graph	0.7150	0.802904

Average across models:

	Accuracy_Mean	ROC_AUC_Mean
0	0.74125	0.848232

3.2 Norm method for TTA

```
[51]: print("Radiomics + demographic + coords Features:")
load_results_table("../Results", ["transformer", "mil", "deep_sets", "graph"],
    use_coords=True, use_demographic=True,
    file_name="tta_norm_center2_aggregated_results.json")
```

Radiomics + demographic + coords Features:

	model	Accuracy_Mean	ROC_AUC_Mean
0	transformer	0.7075	0.811364
1	mil	0.6700	0.741793

2	deep_sets	0.6775	0.753030
3	graph	0.7075	0.784091

Average across models:

	Accuracy_Mean	ROC_AUC_Mean
0	0.690625	0.772569

```
[52]: print("Radiomics + demographic Features:")
load_results_table("../Results", ["transformer", "mil", "deep_sets", "graph"],
    ↪use_coords=False, use_demographic=True,
    ↪file_name="tta_tent_center2_aggregated_results.json")
```

Radiomics + demographic Features:

	model	Accuracy_Mean	ROC_AUC_Mean
0	transformer	0.7050	0.871086
1	mil	0.7550	0.857828
2	deep_sets	0.7375	0.830934
3	graph	0.6800	0.773990

Average across models:

	Accuracy_Mean	ROC_AUC_Mean
0	0.719375	0.83346